Introduction to Business Cycle Data

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November 9, 2022

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- Business cycle fluctuations are costly:
  - Misallocations of capital and labor.
  - Particularly painful for workers that become unemployed and for the families of workers who become unemployed.

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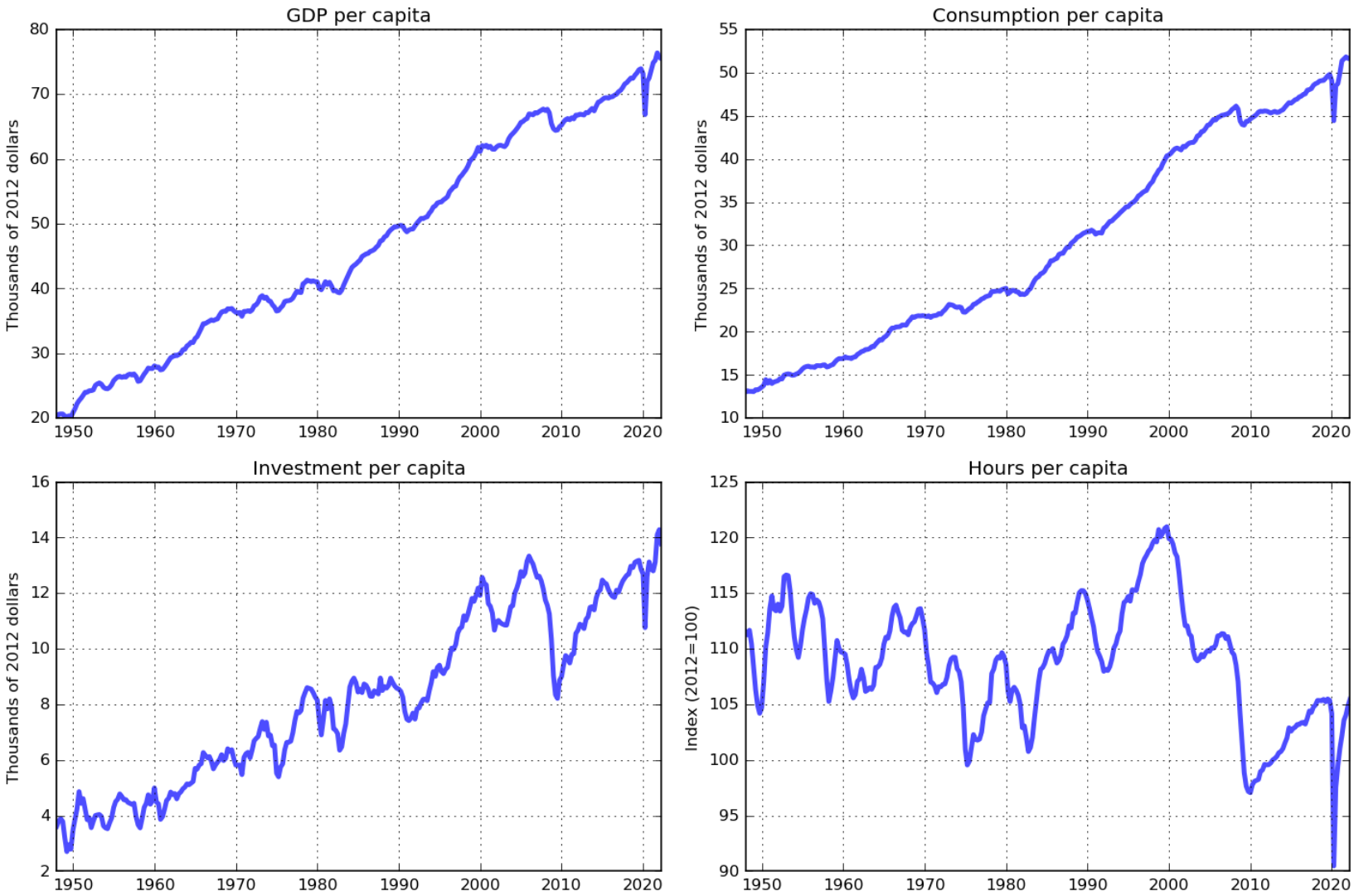
- Both approaches have merits and shortcomings and elements of both are integrated into contemporary business cycle theory.

Figure 1: **GDP, consumption, investment, and hours** for the US from January 1948 to April 2022. Source: FRED.



Trend and Cycle Components

- Suppose that the value of a time series process X_t can be decomposed into two components: a *trend* component and a *cyclical* component.

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- The trend component is the long-run path about which the series fluctuates.

- The cyclical component is the difference between the value of a time series and the trend:

$$X_t^{cycle} = X_t - X_t^{trend} \tag{2}$$

Trend and Cycle Components

- Often, it's useful to express the cyclical component of a time series as the percent deviation of the series from trend (divided by 100):

$$\hat{x}_t = \frac{X_t - X_t^{trend}}{X_t^{trend}} = \frac{X_t^{cycle}}{X_t^{trend}} \tag{3}$$

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- Note:

$$\frac{X_t - X_t^{trend}}{X_t^{trend}} = \frac{X_t^{cycle}}{X_t^{trend}} \approx \log X_t - \log X_t^{trend} \tag{4}$$

Trend and Cycle Components

Example: Percent Deviation from Trend

- Suppose:

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and:

$$\log X_t - \log X_t^{trend} = \log 220 - \log 215 = 0.0230 \tag{8}$$

Figure 2: **GDP, consumption, investment, and hours** per capita for the US from January 1948 to April 2022. Source: FRED.

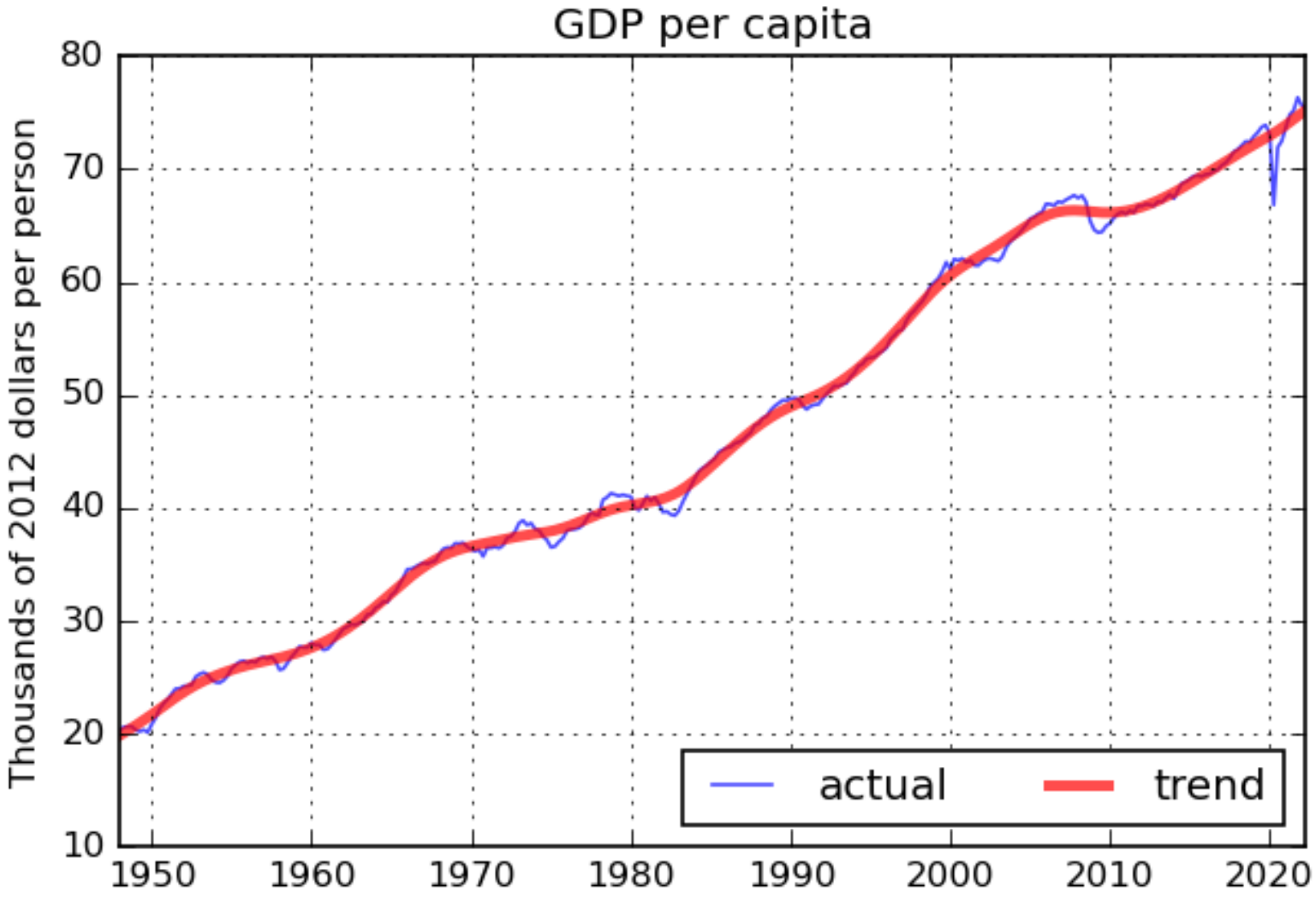

Figure 3: **US GDP per capita:** actual, trend, and cycle from January 1948 to April 2022. Source: FRED.

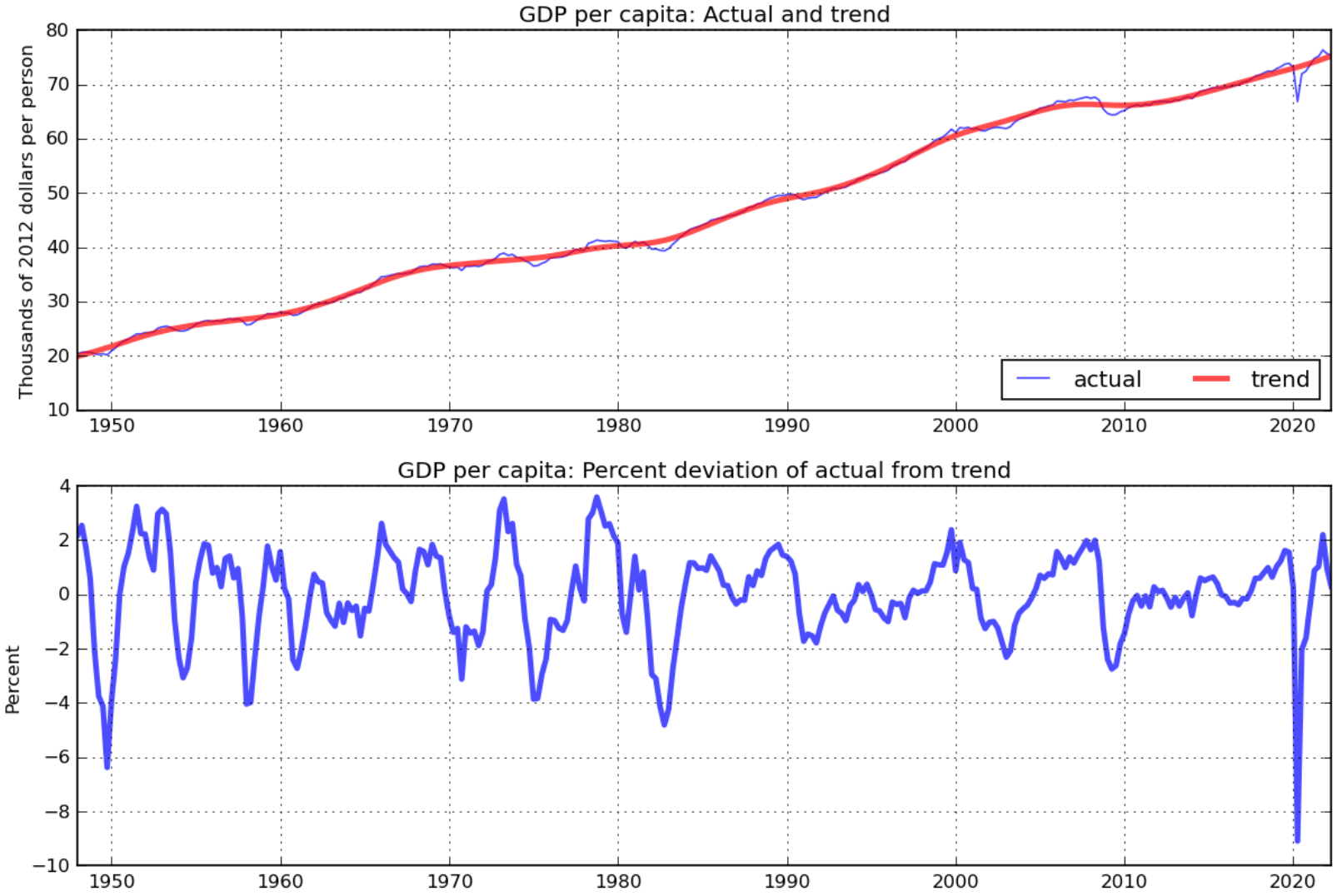

Figure 4: **Business cycle components of GDP, consumption, investment, and hours** for the US from January 1948 to April 2022.

Source: FRED.

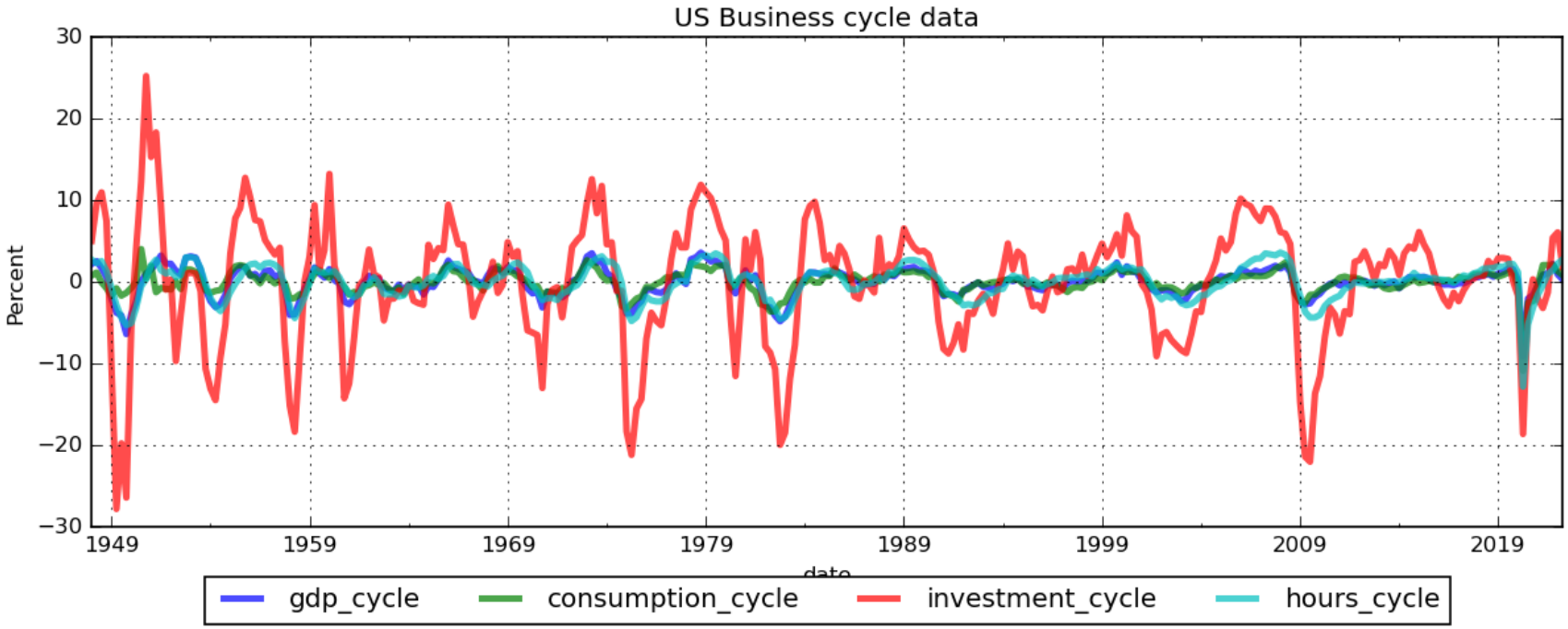

Table 1: **Standard deviations of real business cycle data** from January 1948 to April 2022. Units are percent deviations from trend.

Source: FRED.

GDP	1.688
Consumption	1.337
Investment	7.400
Hours	2.063

Table 2: **Correlations of real business cycle data** from January 1948 to April 2022. Units are percent deviations from trend. Source: FRED.

	GDP	Consumption	Investment	Hours
GDP	1.000	0.814	0.836	0.884
Consumption	0.814	1.000	0.645	0.757
Investment	0.836	0.645	1.000	0.764
Hours	0.884	0.757	0.764	1.000